

Deep Hierarchical Feature Learning: Literature Review and Research Agenda

Scope and bottom line

A source note is necessary at the outset. The exact RTF bibliography you referred to was not exposed by the connected source index in this session. I could retrieve the surrounding project materials: the project manifest explicitly identifies a review of polynomial networks, tensor networks, and tree tensor networks, the recent TTN paper *Benign Loss Landscapes Can Coexist with Worst-Case Hardness*, and work on deep-linear critical loci and loss landscapes as relevant resources. ¹ I therefore reconstructed the bibliography from those materials, the papers identifiable from your description, their citation graphs, and primary papers through September 2026. This matters mainly because there could be papers unique to the RTF that are absent below; the major theoretical lines relevant to your questions are represented.

The high-level conclusion is that **your picture is substantially right, but the field is more advanced in two directions and less advanced in one important sense than your summary suggests.**

First, there is now a fairly coherent explanation for *why hierarchy can create a computational/statistical advantage*: depth can repeatedly turn a feature that is high-dimensional or high-degree in the original representation into a low-dimensional or low-degree feature in a learned representation. This idea appears in several mathematically different forms: hierarchical tensor rank and depth efficiency; staircase/leap complexity; invariance formation in the Random Hierarchy Model; hierarchical-polynomial learning; Dandi-Pesce-Zdeborová-Krzakala's successive dimensionality reduction; and, most explicitly as of 2026, Neural Low-Degree Filtering. ²

Second, the statement that “we do not have mathematical descriptions of full multilayer training dynamics” is now too strong. There are dynamical mean-field theories for jointly trained infinite-width deep networks in feature-learning regimes, adaptive-kernel descriptions of their trained representations, and, in 2026, DMFTs tracking spectral outliers during deep-network training. ³ What is still missing—and this is the crucial version of your claim—is a **Saxe-like, low-dimensional, interpretable solution of full end-to-end training on a genuinely compositional teacher**, where one can read off which internal computations appear, when they appear, their sample thresholds, and the resulting scaling laws. Existing DMFT is much closer to a formal state-evolution framework than to a transparent theory of hierarchical computation.

Third, your TTN proposal has become substantially more compelling because of a very recent result. Furman, Wäldchen, Bei, and Hodgkinson's September 2026 paper argues almost exactly for TTNs as the nonlinear successor to deep linear networks: TTNs contain efficiently evaluable but gradient-hard computations, including arbitrary read-once Boolean formulae; nevertheless their loss landscapes are conditionally benign; and the obstruction to learning can instead reside in **high-order, rank-deficient degenerate saddles**. ⁴ This creates a remarkably clean bridge among target computational structure, hierarchical rank, landscape geometry, and dynamics.

My main recommendation, however, would be to modify your research target slightly. **“Solve TTN training” is not quite the right north star.** The stronger goal is:

Find a compositional teacher–student model for which full simultaneous gradient dynamics closes on a small set of gauge-invariant order parameters, and in which those order parameters describe the sequential formation of internal computation.

TTNs are currently one of the strongest candidates for that model, but the scientifically important object is the **closed dynamical theory of representation formation**, not TTNs themselves.

A useful way to organize what is and is not known is:

Question	Present status
Can depth represent some compositional functions exponentially more efficiently?	Yes, strongly established. ⁵
Can gradient-trained deeper networks learn some hierarchical functions with fewer samples/computation than shallow methods?	Yes, for carefully structured tasks and training protocols. ⁶
Can arbitrarily deep networks provably build hierarchies?	Increasingly yes under layerwise training. Recent 2026 results cover Random Hierarchy Models and polynomial-depth label hierarchies. ⁷
Do we understand the full joint dynamics of a deep nonlinear compositional network?	Not in a Saxe-like, closed, interpretable form. General mean-field/DMFT frameworks exist, but the compositional problem remains open. ⁸
Do we have observables for learned internal structure?	Several promising ones, but no universally accepted closed set: teacher overlaps, representation kernels/spectra, label-weighted moments, AGOP/NFA quantities, tensor ranks/bond spectra, and saddle geometry. ⁹
Can hierarchical feature learning explain empirical neural scaling laws?	Not yet. Fixed-feature spectral theories can produce scaling laws; the missing step is a theory in which the representation spectrum itself evolves hierarchically. ¹⁰

Map of the literature and the strongest results

The literature is easier to understand if one separates **expressivity**, **learnability**, and **training dynamics**. They are often conflated under the heading “advantage of depth,” but they answer different questions.

Depth separation and hierarchical representations. The classical depth-separation literature established that there are functions compactly represented by deep networks that require exponentially larger shallow networks. Eldan and Shamir gave a depth-three versus depth-two separation; Telgarsky constructed

functions requiring exponentially many shallow units; Cohen, Sharir, and Shashua showed that convolutional arithmetic circuits correspond to hierarchical tensor decompositions, with shallow models corresponding to CP-like decompositions and deep ones to hierarchical Tucker decompositions. ¹¹ This tensor perspective is especially relevant to your program: depth is not merely “more nonlinearities”; it changes the **factorization geometry** of the represented high-order coefficient tensor.

These are important foundational results, but they do **not** establish that gradient descent discovers the efficient deep representation. That gap between representability and learnability is exactly why the newer teacher–student literature is more relevant to a theory of pretraining.

Deep linear networks as the canonical dynamical model. Saxe, McClelland, and Ganguli showed that deep linear networks can have nonlinear learning dynamics despite a linear input-output map. Under suitable assumptions, singular modes decouple, yielding exact solutions with long plateaus and rapid transitions; stronger data modes are learned before weaker ones. ¹² Later work sharpened the landscape picture and connected small initialization to rank-increasing saddle dynamics. Jacot et al. conjectured a general saddle-to-saddle picture and proved the first transition in a regime of deep linear networks, while the complete second-order DLN landscape contains both strict and non-strict saddles organized by ranks of partial products. ¹³

Saxe remains the right benchmark for what a successful theory should feel like: there are interpretable latent modes; there are closed equations for them; the equations explain plateaus and transitions; and the statistics of the target determine learning time. The limitation is precisely yours: because the end-to-end function remains linear, DLNs cannot model the phenomenon in which **computation learned at one layer makes a previously inaccessible feature learnable at the next**. Furman et al. make this limitation explicit in motivating TTNs. ⁴

Staircase, merged-staircase, and leap complexity. A conceptually important line that deserves more weight in your current picture is the work of Abbe and collaborators. For Boolean functions, the staircase property formalizes hierarchical reachability of high-order Fourier features from lower-order ones; a layerwise gradient-based algorithm can greedily build higher-order features from previously acquired simpler ones. ¹⁴ The merged-staircase result then identifies a nearly necessary-and-sufficient structural condition for $O(d)$ -scale learning of sparse functions by two-layer mean-field networks, while showing that fixed-feature linear methods cannot achieve the same adaptivity. ¹⁵

The later **leap complexity** theory is particularly close to your ambition. It proposes that for a low-dimensional target embedded in d dimensions, SGD complexity scales roughly as

$$\widetilde{\Theta}\left(d^{\max\{\text{Leap}(f), 2\}}\right),$$

and proves versions of this statement for Gaussian targets and two-layer networks under controlled assumptions. Crucially, the full SGD trajectory sequentially discovers target support through saddle-to-saddle dynamics. ¹⁶ This is already a prototype of the desired triangle

target structure $\longrightarrow$ internal learning sequence $\longrightarrow$ computational scaling.

Its limitation is depth: the hierarchy is largely a hierarchy in the target's low-dimensional support and polynomial/Fourier structure, rather than a truly multilayer learned computational graph.

Three-layer nonlinear feature learning. The three-layer papers are the first rigorous demonstrations that an extra hidden layer can improve *learnability*, not only representation. Nichani, Damian, and Lee prove that layerwise-trained three-layer networks can learn nonlinear features and construct an optimization-based depth separation from two-layer networks. ¹⁷ Wang, Nichani, and Lee subsequently study

$$h(x) = g(p(x)),$$

with p a degree- k polynomial and g degree q . For a broad class of p , a layerwise-trained three-layer network uses $\tilde{O}(d^k)$ samples, whereas kernel methods require $\tilde{\Theta}(d^{kq})$; for quadratic p , their $\tilde{O}(d^2)$ scaling is information-theoretically optimal in their setting. ¹⁸

This makes the central mechanism very concrete: the network does not directly learn the full degree- kq function. It first learns p , thereby converting a high-degree problem in x into an easy low-dimensional problem in the learned coordinate $p(x)$.

The Dandi-Pesce-Zdeborová-Krzakala depth advantage. Your emphasis on this paper is justified. It is probably the cleanest current model of successive hierarchical dimensionality reduction. Their SIGHT teacher has the schematic form

$$f^*(x) = g^* \left(\frac{a^{*\top} P_k(W^*x)}{\sqrt{d^{\varepsilon_1}}} \right),$$

where $x \in \mathbb{R}^d$, $W^*x \in \mathbb{R}^{d^{\varepsilon_1}}$, and successive deep versions introduce dimensions

$$d \rightarrow d^{\varepsilon_1} \rightarrow d^{\varepsilon_2} \rightarrow \dots \rightarrow r, \quad 1 > \varepsilon_1 > \varepsilon_2 > \dots > 0.$$

The paper explicitly interprets depth as repeatedly reducing the effective dimension of the feature-search problem. ¹⁹

For the rigorously treated three-layer case, the first layer recovers the latent subspace W^* using about $\tilde{O}(d^{1+\varepsilon_1})$ samples; the second recovers the nonlinear scalar feature using about $\tilde{O}(d^{k\varepsilon_1})$ samples; and the final readout is essentially finite-dimensional. ¹⁹ So your basic intuition is correct, but I would sharpen it: **the advantage is not generically “another factor of ϵ per layer.”** It is a sequence of task-dependent changes in the exponent of the effective statistical problem, with the bottleneck determined by latent dimension, polynomial degree, and which low-order correlations are available.

There is also a major caveat. The main theorem is considerably further from normal end-to-end GD than the phrase “trained with GD” might suggest. It uses sequential layer training, neuron-wise spherical SGD, independent batches, a correlation loss, reinitialization of the second layer, and a preconditioned single second-layer step. ²⁰ Joint standard backpropagation reproduces the predicted feature-overlap transitions numerically, but that is an empirical extension rather than the theorem. ²¹ Your identification of **full joint dynamics as the missing mathematical object is therefore exactly right.**

There is another caveat that is even more conceptually important: the theorem assumes favorable low-order signal. In particular, the outer function has nonzero first Hermite component and the polynomial feature has nonzero second Hermite component. ²⁰ The authors themselves elevate this into a proposed **Compositional Information Exponent**: every intermediate true feature should retain some low-degree

correlation with the label or a transformation of it. ²¹ That is more fundamental than dimensionality reduction alone. A hierarchy is not useful to GD merely because it exists; the hierarchy must be **locally discoverable** by the gradients.

Random hierarchies and genuinely deep proofs. Cagnetta et al.'s Random Hierarchy Model gives a quite different but complementary picture. Rather than a polynomial teacher on Gaussian inputs, it generates data through a hierarchy of interchangeable “synonyms.” Deep networks succeed by becoming invariant to substitutions within these equivalent lower-level groups; the sample threshold is tied to when correlations between low-level features and labels become statistically detectable. ²² This suggests that the essence of hierarchy may be as much about **quotienting out nuisance degrees of freedom** as about recovering latent subspaces.

Importantly, 2026 has pushed this line beyond merely empirical observations. Ren, Dandi, Krzakala, and Lee prove efficient learning of Random Hierarchy Models by deep convolutional networks under layerwise training, using the principle that intermediate layers can be learned when labels provide sufficiently clean signal and the relevant features are weakly identifiable. ²³ Daniely independently proves that layerwise SGD on residual networks can efficiently learn label hierarchies of polynomial computational depth, moving provable hierarchical learning into genuinely deep models rather than only two- and three-layer networks. ²⁴ Again, however, the tractability comes from **layerwise optimization**.

Tensor factorization and TTNs. The tensor literature contributes three distinct results that should not be collapsed together. First, hierarchical tensor decompositions explain an expressive advantage of compositional depth. ²⁵ Second, gradient descent in hierarchical tensor factorization has an implicit bias toward low hierarchical tensor rank; in corresponding deep convolutional models this can be interpreted as a bias toward locality, with local components activating incrementally. ²⁶ Third, the September 2026 TTN work shows that a mathematically benign landscape can coexist with genuine worst-case computational hardness: non-optimal critical points must be rank-deficient, and parity produces saddles whose escape order grows with problem size. ⁴

That last result is a major conceptual advance for your purposes. It says the right question may not be “why does GD find a global minimum?” The stronger question is:

Why does the target permit a sequence of low-order escape directions that progressively unlock computation?

That question unifies staircase/leap complexity, low-degree compositionality, saddle-to-saddle dynamics, and TTN rank growth.

Finally, the 2026 **Neural Low-Degree Filtering** paper can be read as an attempted synthesis of these themes. In a layerwise vanishing-initialization limit, early feature learning at layer ℓ is governed by low-order label-weighted statistics of the current representation $z_{\ell-1}$,

$$u_\ell = \frac{1}{n} \sum_{\mu} y_{\mu} z_{\ell-1}(x_{\mu}), \quad C_\ell = \frac{1}{n} \sum_{\mu} y_{\mu} z_{\ell-1}(x_{\mu}) z_{\ell-1}(x_{\mu})^{\top}.$$

The network preferentially amplifies leading eigendirections of C_ℓ , applies a nonlinear lift, and then repeats in the new representation. ²⁷ The resulting picture—**filter, nonlinear lift, filter again**—is perhaps the

cleanest present formulation of what “hierarchical feature learning” mathematically means. But the authors are explicit that this is a stylized surrogate derived from an early-time, sequential, vanishing-initialization regime rather than a solution of ordinary end-to-end training. ²⁷

What the mathematics says the mechanism actually is

I would not summarize the literature as having discovered one mechanism. Rather, it has uncovered several mechanisms that increasingly look like different projections of the same underlying process.

Successive reduction of effective search dimension is the mechanism most explicit in the hierarchical polynomial/SIGHT literature. Once W^*x is learned, subsequent optimization operates in d^e rather than d dimensions; once another feature is learned, the dimension can collapse again. ¹⁹ This is probably the cleanest explanation for large improvements in sample-complexity exponents.

Successive reduction of effective degree is a closely related but subtly different description. A target may have very high polynomial degree as a function of the raw input while being degree one or two as a function of the right internal representation. Neural LoFi formalizes the idea that each layer searches only for low-degree correlations in its current coordinates and that nonlinear representation changes make new low-degree correlations available. ²⁸ This is more promising than “dimensionality reduction” as a universal story because representation learning can be valuable even when dimensionality does not literally decrease.

Invariance formation is the Random Hierarchy Model's version of the story. Layers merge equivalent microscopic configurations into common macroscopic features. The reduction is therefore a quotient operation: the representation throws away distinctions that the task declares irrelevant. ²² This may be closer to what language or visual pretraining does than a literal projection onto a low-dimensional linear subspace.

Incremental rank growth is the factorization version. Deep matrix and tensor factorizations tend to activate low-rank components progressively, and hierarchical tensor factorization extends this to hierarchical ranks associated with different cuts of a tree. ²⁹ The attractive feature of this viewpoint is that ranks and singular spectra are intrinsic observables of an internal computation rather than labels attached by the theorist.

Saddle-to-saddle discovery is the dynamical version. In small-initialization regimes, the network spends long periods near lower-complexity invariant manifolds or degenerate saddles; an available correlation eventually produces an unstable/escape direction; training moves to a richer effective model; and the process repeats. This is rigorous in restricted two-layer settings through leap complexity, partially established in DLNs, and has recently been generalized as an explanatory framework across several network architectures. ³⁰

These mechanisms can plausibly be unified by a single schematic process:

current representation	→	detectable task-aligned low-order statistic	→	new feature / rank direction	→	representation
------------------------	---	---	---	------------------------------	---	----------------

This is an inference from the combined literature, rather than an established theorem. It is supported independently by the staircase/leap results, SIGHT/MIGHT, Random Hierarchy Models, hierarchical tensor rank growth, and Neural LoFi. ³¹

This perspective also tells us why **not every compositional function is easy**. A compact compositional circuit can still be invisible to low-order gradient statistics. Parity is the canonical example: it is easy to evaluate with the right circuit but has no useful low-degree correlation from which gradient learning can bootstrap. The TTN result makes this failure geometric: the corresponding optimization landscape can be globally benign and nevertheless possess extraordinarily flat, high-order saddles. ⁴ Thus “the target has a hierarchical representation” is insufficient. One needs a condition closer to:

hierarchical representability + local discoverability of each stage.

The second condition is what staircase structure, leap complexity, local correlation assumptions, compositional information exponents, and Neural LoFi's low-degree criterion are all trying to formalize. ³²

The technical toolkit: Hermite analysis, random matrices, DMFT, and tensor geometry

Your uncertainty about Hermite polynomials and random matrix theory is worth resolving because they play very different roles.

Hermite polynomials are both special and generic. They are special because the exact convenience comes from Gaussian inputs. In one dimension, the Hermite polynomials form the orthogonal polynomial basis of $L^2(\mathcal{N}(0, 1))$. In multiple dimensions, products/Hermite tensors decompose functions into Gaussian polynomial-chaos sectors. The crucial property is that Gaussian correlation acts diagonally by degree: for standardized correlated Gaussian variables X, Y , the degree- k Hermite component propagates with a factor proportional to ρ^k . This makes “first nonzero polynomial degree” an exceptionally clean dynamical quantity. The current hierarchical-learning papers explicitly phrase their assumptions in terms of nonzero Hermite coefficients—for example $\mathbb{E}[g(Z)Z] \neq 0$ and $\mathbb{E}[P_k(Z)\text{He}_2(Z)] \neq 0$. ²⁰ =

Hermes also interact beautifully with Gaussian integration by parts/Stein identities. Derivatives, correlations with projections $w^\top x$, and low-order moments can be converted into one another without producing an uncontrollable hierarchy of statistics. That is a major reason Gaussian teacher–student models are mathematically tractable, not merely a cosmetic choice of basis. Dandi et al.'s proof explicitly uses compositions of Hermite decompositions and asymptotic Gaussianity of intermediate features. ²⁰

But the deeper idea is generic: **decompose the target into irreducible low-complexity modes under the input distribution, and ask which mode has the first nonzero correlation accessible to the learning algorithm**. On the Boolean hypercube this becomes Fourier–Walsh analysis, which underlies staircase and leap complexity; on rotationally symmetric spherical data the analogous objects are spherical harmonics/Gegenbauer sectors. ³³ Thus I would regard Hermite polynomials as the Gaussian coordinate system for a more general spectral principle, not as an accidental technical artifact.

This distinction matters when interpreting information exponents. A theorem that depends quantitatively on the first nonzero Hermite coefficient may be Gaussian-specific. The qualitative statement that gradient

methods preferentially exploit the *lowest-complexity statistically visible component* is much more plausibly universal.

Random matrix theory has a different job. It becomes useful when dimensions, number of samples, and widths are all large enough that the observable of interest is a high-dimensional random matrix: a feature covariance, Gram matrix, Hessian, weight matrix, kernel, or cross-covariance. RMT tells you what part of its spectrum is generic finite-sample “bulk” and what part is task-induced structure.

Dandi et al.'s 2025 RMT work is a particularly clean example. After one aggressive gradient step in a two-layer network, they prove an equivalence to a **spiked random-feature model** and obtain deterministic equivalents for the empirical feature covariance. That lets them calculate both the learned spectrum and exact asymptotic generalization performance. ³⁴ In this setting, RMT is answering questions like:

When does a task-aligned spectral spike emerge from sampling noise?

How large is the spike, and how much does its eigenvector align with the true feature?

How does that spectral transition change test error?

That is extremely relevant to the observables you want.

For hierarchical feature learning, the tantalizing possibility is that **feature formation itself can be formulated as a sequence of spike-emergence transitions**. Neural LoFi already points in this direction: a feature emerges when its low-degree correlation exceeds an empirical-noise floor controlled by the effective dimension of the current representation. ²⁷ A future theory could combine this with RMT so that each new representation has a calculable spectral bulk, and the next internal computation forms when a task-aligned mode exits that bulk.

This would be much closer to a theory of scaling laws than simply proving asymptotic learnability. The sample size at which a spike becomes detectable is itself a **scaling threshold**.

Mean-field theory and DMFT are where I would revise your current understanding most strongly.

Deep feature-learning dynamics are not mathematically unformulated. Bordelon and Pehlevan derive self-consistent DMFT descriptions of representation dynamics in infinite-width deep networks in rich regimes, where the effective NTK becomes time dependent rather than fixed. ³⁵ Lauditi, Bordelon, and Pehlevan further show that feature-learning infinite limits can be represented by task-adaptive kernels, including fixed points arising from DMFT for gradient-flow training with weight decay. ³⁶

Most strikingly for your spectral interests, Lauditi, Pehlevan, and Bordelon's 2026 work develops a two-level DMFT that jointly tracks random spectral bulks and learned outliers, applying it to nonlinear infinite-width networks and deep linear networks. ³⁷ This is almost exactly the mathematical bridge between “dynamics” and “spectral observables” that you are looking for.

The problem is not that DMFT cannot represent full training. The problem is that its order parameters—two-time correlations, responses, effective stochastic processes, evolving kernels—are usually **much less interpretable than Saxe's singular-mode ODEs**, and they have not yet been solved in a way that gives a clean compositional feature-learning sequence for a hierarchical teacher.

So I would formulate the challenge as:

Can one choose a hierarchical teacher/architecture pair for which general DMFT collapses onto a small finite set of feature-overlap and spectral variables?

A TTN's conditional-independence structure may help. But it is not obvious in advance that TTN DMFT will be easier than MLP DMFT; the real tractability will come only if the tree architecture produces an exact or asymptotic **closure of the relevant messages/order parameters**.

There is even recent evidence that DMFT is useful on models simpler than your proposed TTN but richer than static kernels. Nishiyama and Imaizumi derive a low-dimensional DMFT for high-dimensional diagonal linear networks that unifies incremental learning, initialization dependence, convergence rates, and generalization. ³⁸ This suggests an attractive sequence for your program: first reproduce/understand what makes DMFT close in diagonal linear networks; then determine which part survives when scalar products are replaced by tree multilinear contractions.

Tensor geometry provides another, potentially complementary closure principle. A TTN possesses natural ranks and singular spectra associated with every internal tree edge. These are coordinate-independent properties of the represented tensor, unlike individual weights, which are subject to gauge transformations along bonds. Furman et al. show that rank deficiency is not just a representational detail: all non-global critical points in their setting are rank deficient, and the minimum-norm/balanced condition is related to the gauge geometry and is preserved by gradient flow. ⁴ That makes hierarchical rank variables unusually promising observables.

The methodological lesson is that **you probably want both statistical-physics and algebraic/tensor tools**:

tensor geometry $\Rightarrow$ what invariant structure exists,
DMFT/RMT $\Rightarrow$ how noisy high-dimensional training moves that structure.

Training dynamics and the observables that seem most promising

A useful distinction is between four levels of “understanding the dynamics.”

At the strongest level, one has a low-dimensional exact reduction such as Saxe's singular-mode equations.

¹² Below that, one can have an asymptotically exact state evolution/DMFT whose solution still requires coupled integral equations. ³⁹ A third level consists of controlled training protocols deliberately constructed to expose the desired mechanism, as in the three-layer and SIGHT/MIGHT results. ⁴⁰ The weakest—but still scientifically valuable—level is a mechanistic surrogate such as Neural LoFi that predicts some early learned features of ordinary backpropagation without claiming to reproduce its whole trajectory. ²⁷

The main gap is between the first two levels and the third: **we do not have a simple state evolution for simultaneously trained deep nonlinear layers that visibly performs recursive feature discovery.**

For your longer-term pretraining program, the choice of observables may be as important as the choice of solvable model. I would impose four requirements on a useful observable:

gauge/parameterization invariant, dynamically predictable, structurally interpretable, connected to

By that criterion, I would prioritize the following families.

Teacher–student overlap matrices are the gold standard for toy models. Dandi et al. explicitly measure first-layer subspace overlap and hidden-feature overlap,

$$M_W \sim W_1 W^{\star\top}, \quad M_h \sim \mathbb{E}[h(x)h^{\star}(x)^{\top}],$$

and observe sharp feature-learning transitions as sample complexity is varied. ²¹ In a TTN teacher–student model, there should be an analogous overlap for every internal edge/subtree. These are maximally interpretable because the true internal computation is known.

Their drawback is equally clear: real pretraining has no known teacher representation. So they are ideal **development order parameters**, but not final empirical observables.

Label-weighted low-order moment spectra are, in my view, the most promising teacher-free mechanistic observable. At each layer, Neural LoFi uses $E[yz]$ and $E[yzz^{\top}]$, or their residualized/multiclass analogues. ²⁷ They directly measure which feature combinations are currently visible to a low-order gradient signal. Tracking their leading eigenvalues during training could tell you when a representation has made the next computation accessible.

This suggests a powerful experimental diagnostic: at checkpoint t and layer ℓ , compute the spectrum of

$$C_{\ell,t} = \mathbb{E}[r_t(x) z_{\ell,t}(x) z_{\ell,t}(x)^{\top}],$$

where r_t is an appropriate residual/error signal rather than the raw label. A theory of pretraining should ideally predict when an eigenvalue separates, what feature its eigenvector represents, and how that event changes the corresponding operator one layer deeper.

Representation Gram matrices and their spectra are useful, but insufficient alone. Activation kernels

$$K_{\ell}(x, x') = \frac{1}{p_{\ell}} z_{\ell}(x)^{\top} z_{\ell}(x')$$

tell you about effective dimension, clustering, invariance, and spectral concentration. Adaptive-kernel and DMFT approaches make these quantities natural state variables. ⁸ RMT can potentially separate task-induced directions from finite-sample spectral bulk. ⁴¹ But an unsupervised Gram spectrum can change dramatically without telling you *what computation* the network has learned; it should therefore be paired with a task-conditioned statistic.

The Neural Feature Ansatz deserves attention, but not the status of a universal law. The original Deep Neural Feature Ansatz proposes that learned weight-Gram matrices—the neural feature matrices—align with average gradient outer products,

$$\text{AGOP} = \mathbb{E}_x [\nabla_x f(x) \nabla_x f(x)^\top],$$

or analogous layerwise quantities. ⁴² Follow-up work shows that this correlation can be understood as alignment between the singular structure of weights and preactivation tangent features, and derives early-time alignment from SGD statistics. ⁴³

That makes the NFA family compelling because gradients provide a direct measure of “which directions the model’s computation is sensitive to.” But your current wording should be sharpened: it is not simply that the **Gram matrix of weights equals the Gram matrix of gradients on examples**. The relevant objects live in feature/input coordinate space: roughly $W^\top W$ is compared with an average outer product of derivatives with respect to the preceding representation.

More importantly, a 2026 analysis proves the NFA exactly in balanced deep *linear* networks—with a depth-dependent exponent $1/L$ —but also constructs counterexamples for nonlinear architectures even when those networks fit the data well. ⁴⁴ I would therefore treat NFA alignment as an **observable and phenomenological hypothesis**, not the foundation of the theory.

TTN bond spectra may be the best model-specific observable available. Cut any internal TTN edge and matricize the represented tensor across that bipartition. Its singular values tell you the effective hierarchical rank across that computational interface. In a teacher–student experiment, one can monitor:

$$\{s_{e,j}(t)\}_j \quad \text{for every edge } e,$$

together with student–teacher singular-subspace overlaps. Furman et al.’s connection between non-optimal critical points and rank deficiency gives these quantities direct optimization significance. ⁴ Hierarchical tensor factorization’s low-rank implicit bias gives them dynamical significance as well. ²⁶

This is one place where TTNs are arguably superior to MLPs as a theory model: they come with a **canonical hierarchy of intrinsic rank observables**.

Finally, I would explicitly monitor **saddle degeneracy/escape order**. Hessian eigenvalues alone will miss precisely the phenomena Furman et al. identify, because a high-order degenerate saddle can have no informative negative Hessian direction. ⁴ For simple TTNs it may be feasible to calculate the lowest nonzero derivative of the loss transverse to a rank-deficient stratum. That number could turn out to be the dynamical analogue of information exponent or leap complexity.

This suggests a speculative but attractive correspondence:

$$\boxed{\text{low-degree statistical signal} \longleftrightarrow \text{low-order saddle escape} \longleftrightarrow \text{fast feature emergence}}$$

and conversely

$$\boxed{\text{high information/leap degree} \longleftrightarrow \text{high-order flat saddle} \longleftrightarrow \text{large computational/sample exponent.}}$$

The staircase/leap literature and the TTN parity result independently support pieces of this correspondence, but proving it generally would be a substantial new result. ⁴⁵

Critique of your current understanding and an improved TTN program

Your current synthesis is unusually close to what I would take away from this literature. The most useful corrections are about **scope and strength of the results**.

Your statement that hierarchical feature learning is central to the advantage of depth is defensible as a research hypothesis, but I would avoid presenting it as established in full generality. Classical depth separations prove representational advantages; modern teacher–student results prove learning advantages on specific compositional target classes; Random Hierarchy Models support a related mechanism empirically and increasingly theoretically. ⁴⁶ What is not established is that the success of large-scale pretrained Transformers, for example, is quantitatively explained by the same mechanism.

Your reading of the Dandi–Pesce–Zdeborová–Krzakala paper is basically correct. The important refinement is that their deepest mathematical idea is arguably not just “each layer reduces dimensionality,” but **each learned representation makes the next target feature accessible through a low-order correlation**. Their own Compositional Information Exponent conjecture points in this direction. ²¹ That distinction will matter enormously outside Gaussian polynomial tasks.

Your statement that a three-layer network can beat a two-layer network is correct in several precise settings, but should always carry the qualifiers “for specially structured targets” and usually “under layerwise/controlled optimization.” ⁴⁷ There is no generic theorem saying that adding a third layer lowers sample complexity.

I would also modify the statement that tensor-network papers show saddle-to-saddle dynamics. Hierarchical tensor-factorization work establishes incremental low-rank/locality bias, while exact saddle-to-saddle descriptions are cleaner in diagonal/deep linear settings and in the leap-complexity work. ⁴⁸ The new TTN result does something slightly different and perhaps more useful: it geometrically identifies **where hard dynamics must live**—rank-deficient, highly degenerate saddles—without yet solving the trajectories between them. ⁴

On Hermite polynomials, your intuition was right: their exact utility is strongly tied to Gaussianity. But they instantiate a general spectral philosophy rather than a Gaussian curiosity. The analogous bases change with the data distribution/symmetry, while “lowest visible degree controls learning difficulty” survives across Gaussian and Boolean results. ⁴⁹

On RMT, I would elevate its importance but narrow its conceptual role. RMT is **not an explanation of hierarchy**. It is potentially the theory of the *measurement layer*: given noisy high-dimensional features, it tells you whether a learned direction is statistically detectable, how feature spectra evolve, and when signal separates from noise. ⁵⁰ That makes it a natural technology for converting a feature-learning mechanism into quantitative sample thresholds and, ultimately, scaling laws.

On full dynamics, your diagnosis should be rewritten from

“We have no mathematical analysis of full joint training”

to

“We lack an interpretable closed theory of full joint training on a nonlinear hierarchical teacher.”

That formulation survives the existence of deep-network mean-field/DMFT theories and identifies the actual bottleneck. 8

How I would modify the TTN research program

The strongest argument for TTNs is now quite concrete. They strictly generalize the chain topology of DLNs to branching multilinear computation, support efficiently evaluable nonlinear hierarchical functions, have natural rank observables, and—as of September 2026—are known to simultaneously admit benign global landscape properties and computationally hard targets. 4 That is an unusually good list of desiderata for a deep-learning toy model.

But **bond dimension one should be viewed as a calibration model, not the main scientific model**. A bond of dimension one carries only a scalar across that tree edge. It can still represent high-degree multiplicative functions, so the dynamics need not be trivial, but it has no nontrivial competition among edge subspaces and no meaningful hierarchical-rank growth. The rank phenomena that make TTNs interesting require at least several bond directions. This follows directly from the TTN definition in which bond dimension is the dimension of the intermediate vector space and from the role of rank deficiency in the landscape theory. 4

So I would pursue the following ladder.

Start with the bond-one model, but ask a narrow question. Derive the exact scalar gradient-flow equations on a balanced tree. Identify conserved balancedness quantities, characterize the origin and other degenerate saddles, calculate escape times as initialization tends to zero, and determine how those times scale with tree depth. This gives you the TTN analogue of the scalar/diagonal DLN calculation.

The key deliverable here is not “hierarchical feature learning”—the model is too narrow for that—but a clean understanding of how **branching changes multiplicative gradient dynamics**.

Then move immediately to diagonally decomposable or orthogonally decomposable TTNs with bond dimension $r > 1$. This, in my view, is the real analogue of the Saxe model. Choose teacher cores that share compatible orthogonal bases so that the represented tensor has a hierarchical analogue of an SVD. Initialize the student in, or near, the same symmetry class. Then ask whether full gradient flow preserves enough of that structure to reduce training to amplitudes

$$a_{e,j}(t),$$

one per edge e and hierarchical mode j .

If this works, you could obtain the first genuinely nonlinear version of Saxe's program:

teacher hierarchical spectrum $\rightarrow$ closed mode ODEs $\rightarrow$ ordered feature acquisition.

This would be a very strong result even before finite-data effects.

The next step should be a planted low-hierarchical-rank teacher with a wider, randomly initialized student. At this point exact solvability will probably disappear. Take a balanced depth- L tree, bond dimension $R \rightarrow \infty$, fixed teacher rank r , and inputs generated either from Gaussian leaf variables or another product distribution. The obvious candidate order parameters are edge-wise student-teacher overlaps,

$$Q_e(t),$$

student self-overlaps,

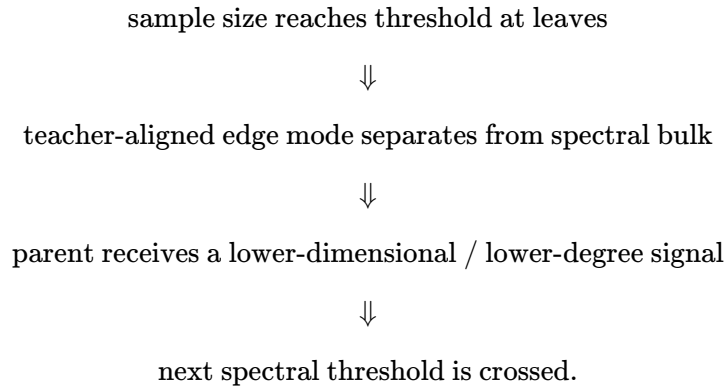
$$C_e(t),$$

and perhaps response functions $R_e(t, t')$. The tree may permit recursive cavity/message-passing equations because disjoint subtrees are conditionally independent before they meet at a parent.

This is where I would try DMFT. The objective should not be to derive the most general possible TTN DMFT. It should be to discover a scaling in which the DMFT **closes recursively along the tree**, ideally reducing two-time functional order parameters to a few edge-wise scalar or matrix quantities.

Finite-sample effects should be added only after the population dynamics are understood. Then RMT becomes valuable. Every edge has a sample-dependent cross-covariance or feature covariance. The natural question becomes whether teacher-aligned modes appear as spectral outliers from a random bulk. The one-step RMT work shows that such learned-feature spectra can be characterized sharply in simpler networks. ⁴¹

That yields a plausible recursive mechanism:



That is the version of the TTN program that could genuinely produce scaling laws.

You should deliberately include both easy and hard teachers. Easy teachers should satisfy an analogue of compositional low-degree/staircase structure. Hard teachers should include parity/read-once formulas for which Furman et al. predict severe saddle degeneracy. ⁴ Otherwise a model that is always learnable risks reproducing the weakness of DLNs: it cannot explain why realistic structure matters.

The deepest possible result would be a theorem that assigns every internal node or edge a local discoverability exponent q_e , then derives a training/sample exponent recursively from the tree. One might hope for something schematic like

$$\kappa_{\text{learn}} = \max_{e \in E} \Phi(q_e, d_e, \kappa_{\text{children}(e)}),$$

where q_e measures the first useful low-order correlation/saddle escape and d_e the effective dimension entering that node. This formula is speculative, but it would unify the best ideas in Dandi's dimension exponents, leap complexity, and TTN saddle geometry. ⁵¹

One danger deserves emphasis: a matched TTN student may **bake in too much of the teacher's hierarchy**. If both teacher and student have the same known tree, success could come from architectural supervision rather than learned hierarchical computation. You therefore eventually need mismatched experiments: permuted trees, overcomplete trees, weakly connected alternatives, or bilinear MLPs that can represent the same tensor without being handed its exact factorization. Furman et al. note connections between bilinear neural networks and TTN-like computation, which offers a natural route for this transfer test. ⁴

Prioritized open problems and the path toward a theory of pretraining

The following are the questions I would rank highest. The ordering reflects not just intrinsic importance but how much solving each one would unlock the rest of the program.

1. Obtain a closed theory of full, simultaneous gradient dynamics for a genuinely compositional deep teacher.

This is the central bottleneck. Layerwise proofs now exist for three-layer polynomial models, deep Random Hierarchy Models, and polynomial-depth hierarchical models; full deep-network DMFT exists at a formal level; what is missing is their intersection. ⁵²

A successful solution should show, without freezing or manually sequencing layers, how simultaneous gradient flow nevertheless produces a sequence such as

feature A $\rightarrow$ feature B becomes learnable $\rightarrow$ feature B $\rightarrow$ feature C becomes learnable.

TTNs with diagonal/orthogonally decomposable cores are an excellent candidate. I would prioritize this over attempting a completely general large-bond TTN DMFT immediately.

2. Identify the correct necessary-and-sufficient notion of "hierarchical learnability."

This is conceptually even deeper than dimensionality reduction. The literature currently contains several candidate notions:

staircase/merged staircase, leap complexity, local correlation, information exponent, composition

They all formalize, in different languages, whether the next component of a computation becomes visible after previous components have been learned. ⁵³

A major theoretical result would establish equivalences or inequalities among these notions in a common model. For example: does low compositional information exponent imply low saddle escape order? Does a TTN version of staircase structure guarantee polynomial escape from all rank-deficient saddles? Does leap complexity become a tree-local quantity?

This is likely where a genuine theory of **which computations pretraining can discover** will come from.

3. Find a small, gauge-invariant set of observables that closes approximately under training.

For a theory of pretraining, explaining the final predictor is not enough. One needs observables that can be measured throughout training and that predict the next structural transition.

My proposed core set for a TTN or related model would be

edge hierarchical singular spectra,
student-teacher feature/subspace overlaps,
label/residual-weighted low-degree moment spectra,
AGOP / tangent-feature alignment,
representation effective dimensions,
local saddle escape order.

The first two are ideal for theory, the middle three transfer naturally to ordinary networks, and the final one connects dynamics to computational hardness. Existing work supplies pieces of this package, but nobody has shown that these quantities jointly form an approximately closed dynamical system. ⁵⁴

I would consider **closure** the decisive criterion. An observable that beautifully describes a trained network but cannot predict its future trajectory is less useful for your goal than a cruder observable with a state-evolution equation.

4. Derive scaling laws from representation formation rather than from a fixed kernel.

There is already a substantial theory relating power-law eigenspectra of fixed kernels/random features to power-law generalization curves. Recent 2026 work derives dynamical neural scaling laws in random-feature regression through DMFT and power-law kernel spectra. ⁵⁵ But that theory treats the feature space as given.

Your desired theory needs to explain why the relevant spectrum itself emerges during training.

A very appealing picture is a succession of detectability transitions:

$$n < n_1 : \text{feature 1 buried in noise,}$$

$n_1 < n < n_2$: feature 1 learned, feature 2 inaccessible,

$n_2 < n < n_3$: feature 2 learned, ...

The SIGHT experiments already show sharp thresholds in $\kappa = \log n / \log d$, and Neural LoFi explicitly proposes a feature-emergence criterion based on population correlation exceeding an effective-dimension-dependent empirical noise floor. ⁵⁶

The scaling-law problem is therefore not necessarily independent of hierarchical feature learning. It may be:

scaling law = aggregate envelope of many feature-emergence thresholds.

That is a hypothesis, not a current theorem, but I think it is one of the most promising bridges from this literature to pretraining.

5. Explain plateaus and emergence by connecting target complexity to degenerate saddle geometry.

Furman et al.'s parity result gives a remarkable proof of concept: a computationally hard target corresponds to a saddle whose escape order grows with target size, even though the landscape contains no relevant bad local minima. ⁴ Leap-complexity results independently connect target structure to saddle-to-saddle training time. ¹⁶

A general theorem relating

statistical degree / leap $\leftrightarrow$ saddle degeneracy $\leftrightarrow$ plateau duration

would be extraordinarily valuable. It could provide a mathematical language for “emergent capabilities”: training appears static because all low-order derivatives in the relevant direction vanish, then a new computation becomes active once the system has moved onto a stratum where a lower-order escape exists.

TTNs are unusually well suited to this because their critical strata can be described through rank deficiencies.

6. Establish universality beyond Gaussian inputs and Hermite analysis.

A theory of pretraining cannot ultimately depend on exact Gaussian rotational invariance. The right target is to understand which parts of the Gaussian calculations survive for non-Gaussian product distributions, anisotropic inputs, correlated latent variables, and eventually data-manifold distributions.

The likely invariant content is not “Hermite coefficient number two” but something like:

the smallest complexity sector of the current representation that has statistically detectable correlation with the residual objective.

The staircase/leap results on Boolean inputs and Gaussian information-exponent results already indicate that such a distribution-independent abstraction should exist. ⁵⁷

A useful intermediate goal would be to prove the same hierarchical dynamics under several different input ensembles and show that only a small collection of low-order moments/spectral properties changes. That would distinguish genuine mechanism from Gaussian algebra.

7. Move from supervised teacher-student learning to pretraining objectives.

This is ultimately the largest conceptual jump. Current hierarchical feature-learning theory generally assumes a scalar supervised target whose correlations tell the network what features to build. Pretraining has a vastly richer supervision signal: next-token prediction, masked prediction, contrastive objectives, or generative denoising expose many latent variables simultaneously.

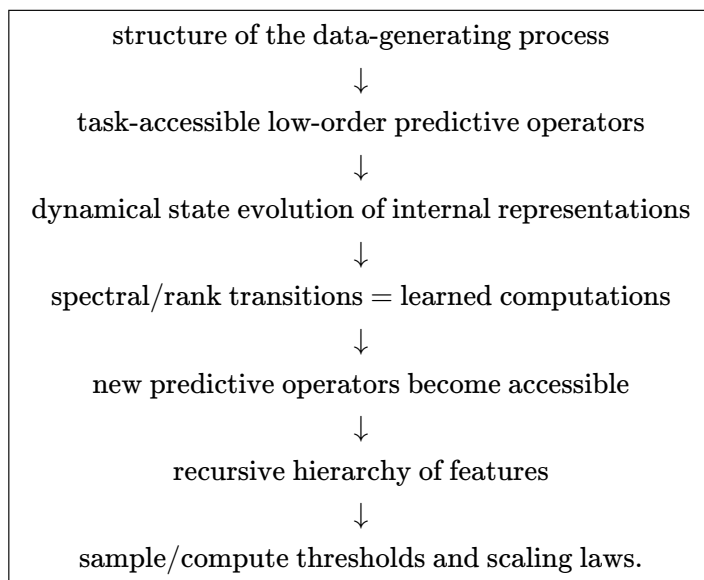
A good theory therefore needs an analogue of the scalar label-weighted moment

$$\mathbb{E}[y z z^\top]$$

for a high-dimensional predictive objective. In language modeling, for example, the “label” is an entire conditional distribution. The relevant feature-learning operator may involve covariance between a representation and the **prediction residual vector**, Fisher/gradient covariances, or singular modes of a conditional cross-covariance operator.

This is one reason I would not make the Neural Feature Ansatz itself the foundation. A better target is a general theory in which AGOP, residual-weighted moments, adaptive kernels, and TTN bond spectra emerge as different projections of a common predictive-information operator. Existing adaptive-kernel, NFA, and Neural LoFi results provide partial evidence for such an interpretation, but none gives it in the form needed for modern pretraining. ⁵⁸

Putting these pieces together, the theory of pretraining you are aiming for could plausibly have the following architecture:



The strongest existing work already supplies almost every box individually: Saxe supplies solvable nonlinear-in-parameters dynamics; staircase and leap supply target-dependent computational hierarchy; hierarchical-polynomial and SIGHT/MIGHT results supply dimensionality reduction and sample-complexity gains; Random Hierarchy Models supply invariance-building and more natural hierarchical data; RMT supplies signal-versus-noise spectral thresholds; DMFT supplies a framework for full rich-regime deep dynamics; Neural LoFi supplies an explicit iterative low-degree mechanism; and TTNs now supply a nonlinear compositional surrogate with intrinsic hierarchical ranks, hard functions, and analyzable degenerate saddles. ⁵⁹

What the field still lacks is the **single model in which those boxes are simultaneously connected by equations**. That, rather than another depth-separation result, is the most important missing piece. A carefully chosen TTN teacher–student model—starting with diagonal/orthogonally decomposable cores rather than stopping at bond dimension one—looks like one of the most credible routes to obtaining it.

1 SOURCE_MANIFEST.md

<https://drive.google.com/file/d/1XFMLjiTDIE9hqvWdtawqE7ATjNPK1ER/view?usp=drivesdk>

2 ²⁵ ⁴⁶ <https://proceedings.mlr.press/v49/cohen16.html>

<https://proceedings.mlr.press/v49/cohen16.html>

3 ⁸ ³⁵ ³⁹ <https://arxiv.org/abs/2210.02157>

<https://arxiv.org/abs/2210.02157>

4 <https://arxiv.org/abs/2609.13057>

<https://arxiv.org/abs/2609.13057>

5 <https://arxiv.org/abs/1602.04485>

<https://arxiv.org/abs/1602.04485>

6 ¹⁸ ⁴⁰ ⁴⁷ <https://arxiv.org/abs/2311.13774>

<https://arxiv.org/abs/2311.13774>

7 ²³ <https://arxiv.org/abs/2601.19756>

<https://arxiv.org/abs/2601.19756>

9 ¹⁹ ²⁰ ²¹ ⁴⁹ ⁵² ⁵⁴ ⁵⁶ <https://arxiv.org/html/2502.13961v4>

<https://arxiv.org/html/2502.13961v4>

10 ⁵⁵ <https://proceedings.mlr.press/v306/kramp26a.html>

<https://proceedings.mlr.press/v306/kramp26a.html>

11 <https://proceedings.mlr.press/v49/eldan16.html>

<https://proceedings.mlr.press/v49/eldan16.html>

12 ⁵⁹ <https://arxiv.org/abs/1312.6120>

<https://arxiv.org/abs/1312.6120>

13 <https://arxiv.org/abs/2106.15933>

<https://arxiv.org/abs/2106.15933>

14 ³¹ ³² ³³ ⁵³ ⁵⁷ <https://arxiv.org/abs/2108.10573>

<https://arxiv.org/abs/2108.10573>

15 <https://arxiv.org/abs/2202.08658>

<https://arxiv.org/abs/2202.08658>

16 30 45 51 <https://arxiv.org/abs/2302.11055>

<https://arxiv.org/abs/2302.11055>

17 <https://arxiv.org/abs/2305.06986>

<https://arxiv.org/abs/2305.06986>

22 <https://journals.aps.org/prx/abstract/10.1103/PhysRevX.14.031001>

<https://journals.aps.org/prx/abstract/10.1103/PhysRevX.14.031001>

24 <https://arxiv.org/abs/2601.00455>

<https://arxiv.org/abs/2601.00455>

26 29 48 <https://arxiv.org/pdf/2201.11729v4.pdf>

<https://arxiv.org/pdf/2201.11729v4.pdf>

27 28 <https://arxiv.org/abs/2605.13612>

<https://arxiv.org/abs/2605.13612>

34 <https://arxiv.org/abs/2410.18938>

<https://arxiv.org/abs/2410.18938>

36 58 <https://proceedings.mlr.press/v267/lauditi25a.html>

<https://proceedings.mlr.press/v267/lauditi25a.html>

37 <https://arxiv.org/abs/2605.07870>

<https://arxiv.org/abs/2605.07870>

38 <https://proceedings.mlr.press/v300/nishiyama26a.html>

<https://proceedings.mlr.press/v300/nishiyama26a.html>

41 50 <https://proceedings.mlr.press/v258/dandi25a.html>

<https://proceedings.mlr.press/v258/dandi25a.html>

42 <https://arxiv.org/abs/2212.13881>

<https://arxiv.org/abs/2212.13881>

43 <https://arxiv.org/abs/2402.05271>

<https://arxiv.org/abs/2402.05271>

44 <https://arxiv.org/abs/2510.15563>

<https://arxiv.org/abs/2510.15563>